

MA 576 : Nonlinear Optimization

Course Code: MA 576

Course Name: Nonlinear Optimization

L-T-P-C: 3-0-2-4

Students intended for: B. Tech/ MS/PhD/MSc

Elective or Core: Elective

Prerequisites: Probability and Statistics, Linear Algebra, and Optimizations

Mutual Exclusion: DS 413, CS 669

Approval: 59th BoA

Course Contents

- **Fundamentals and Convexity:** Introduction and Motivation: Nonlinear Optimization Problems, Convex Sets and Convex Functions: Definition and properties, Jensen's inequality, weaker forms of convexity and relationship between different forms of convexity. [6 Lectures]
- **Unconstrained Optimization Theory:** Optimality Conditions: First-order necessary conditions, Second-order necessary and sufficient conditions, Sub-gradients and Non-smooth Convex Optimization: Definition and properties of sub-gradients, Optimality conditions using sub-gradients, Introduction to optimization of non-smooth convex functions. [8 Lectures]
- **Unconstrained Optimization Algorithms:** Line Search Methods: Exact and Inexact line search methods, Gradient Descent Methods: Steepest descent method and convergence, Variants of gradient descent, Newton's Method: Algorithm and motivation, Local quadratic convergence analysis, Quasi-Newton Methods: Motivation and basic principles, BFGS algorithm. [12 Lectures]
- **Constrained Optimization Theory:** Lagrange multipliers, Slater's and Linearly Independent Constraint Qualifications, Karush-Kuhn-Tucker necessary and sufficient conditions: derivation and interpretation, Inequality Constrained Optimization: Active and inactive constraints, Duality in Nonlinear Programming: Lagrangian dual function, weak and strong duality theorems. [8 Lectures]
- **Constrained Optimization Algorithms:** Exterior penalty methods, Interior point methods, Practical aspects and choice of penalty/barrier parameters; Augmented Lagrangian Methods: Motivation for combining penalty and Lagrangian approaches. [8 Lectures]

Laboratory/practical/tutorial Modules:

- This course will also include one tutorial session every week dedicated to solving exercises relevant to the studied topics.

Textbooks:

1. Oliver Stein, **Basic Concepts of Nonlinear Optimization**, Springer, 2024.
2. Nocedal, J. & Wright, S. J., **Numerical Optimization**, 2nd Edition, Springer, 2006.
3. Boyd, S. & Vandenberghe, L., **Convex Optimization**, Cambridge University Press, 2004.



Reference Books:

1. Antoniou, A. & Lu, W.-S., **Practical Optimization: Algorithms and Engineering Applications**, Springer, 2007.
2. Bazaraa, M. S., Sherali, H. D., & Shetty, C. M., **Nonlinear Programming: Theory and Algorithms**, 3rd Edition, John Wiley & Sons, 2006.
3. H. A. Eiselt and Carl-Louis Sandblom, **Nonlinear Optimization Methods and Applications**, Springer, 2020.